

Your grade: 100%

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Next item →

1.

Can an MPI blocking send be part of a broadcast communication?

1 / 1 point
- Yes; the send returns once all receiving ranks call their blocking receives.

No; blocking sends and receives must occur pairwise

Nice work

Correct.
2.

What is the most important step in avoiding deadlock in an MPI application?

1 / 1 point
- Ensuring the message size isn't too large - MPI will prevent messages of a certain size from being sent in their entirety to keep message bandwidth open, which can cause deadlock

Ensuring each send is matched pairwise with a receive, so that different ranks aren't waiting for messages from each other at the same time

Ensuring the number of ranks doesn't exceed the number of cores in the node on which the program is running

All of the above

Nice work

Correct. MPI send blocks until the message is transmitted. It is therefore crucial for the destination rank to be prepared to receive in order to avoid a deadlock.
3.

Which of the following would be a good use of an MPI barrier?

1 / 1 point
- Synchronizing file access - preventing ranks from reading from a particular file until all ranks have written their data to it

Synchronizing program flow - calling a barrier directly after a receive across all ranks, so that the program waits until each rank receives data from each other

None of the above - MPI barriers were deprecated in MPI version 3, because synchronous calls are often more effective

Both A and B

Nice work

Correct.
4.

What is the main purpose of MPI communicators?

1 / 1 point
- To initialize MPI processes

To handle a group of processors and their interrelationships expressed by MPI rank

To optimize network topology

To manage memory allocation

Nice work

Correct. This defines the core functionality of communicators in MPI.
5.

Which of the following is NOT a requirement when creating custom communicators?

1 / 1 point
- The mapping between processors and ranks must be one-to-one

Processes can belong to multiple communicators

Communicators must have an equal number of processes

Communicators are logical constructs

Nice work

Correct. Communicators can have varying numbers of processes.
6.

What is the purpose of the 'color' parameter when creating a new communicator?

1 / 1 point
- To determine the rank in the new communicator

To specify the number of processes

To determine group membership

To set the communication speed

Nice work

Correct. Color is used to categorize processes into groups.
7.

Which MPI derived datatype allows for multiple consecutive elements of the same type?

1 / 1 point
- MPI_Type_vector

MPI_Type_indexed

MPI_Type_contiguous

Nice work

Correct. Contiguous type represents consecutive elements of the same type.
8.

What is a key advantage of using MPI derived datatypes?

1 / 1 point
- They automatically parallelize code

They increase data transmission speed

They can simplify communication by eliminating temporary buffers

They reduce memory usage

Nice work

Correct. This is a primary benefit of derived datatypes.
9.

Which derived datatype allows for non-contiguous elements with variable strides and block sizes of different types?

1 / 1 point
- MPI_Type_contiguous

MPI_Type_vector

MPI_Type_indexed

MPI_Type_struct

Nice work

Correct. Struct type offers the most flexibility in data structure definition.
10.

What is the purpose of 'committing' a derived datatype?

1 / 1 point
- To allocate memory for the datatype

To make it accessible to the MPI library

To optimize its performance

To link it with other datatypes

Nice work

Correct, committing registers the datatype with the MPI system.
11.

In the context of 2D domain decomposition, why are MPI derived types particularly useful?

1 / 1 point
- They increase computation speed

They allow for automatic load balancing

They can make ghost zone exchanges more efficient regardless of the programming language

They reduce the number of dimensions to 1D

Nice work

Correct, they help manage non-contiguous memory access efficiently.
12.

What is the main difference between MPI_Type_vector and MPI_Type_indexed?

1 / 1 point
- Vector type allows for different element types, while indexed does not

Indexed type allows for irregular spacing and variable block sizes, while vector does not

Vector type is faster than indexed type

Indexed type requires less memory than vector type

Nice work

Correct, this highlights the key distinction between these types.
13.

When creating an MPI_Type_vector, what three main parameters are needed?

1 / 1 point
- Number of blocks, elements per block, and stride between blocks

Color, key, and group size

Data type, memory address, and total size

Start index, end index, and step size

Nice work

Correct, these parameters define the structure of a vector type.